Competing Risks and Time-Dependent Covariates

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Time-dependent covariates are frequently encountered in regression analysis for event history data and competing risks. They are often essential predictors, which cannot be substituted by time-fixed covariates. This study briefly recalls the different types of time-dependent covariates, as classified by Kalbfleisch and Prentice [*The Statistical Analysis of Failure Time Data*, Wiley, New York, 2002] with the intent of clarifying their role and emphasizing the limitations in standard survival models and in the competing risks setting. If random (internal) time-dependent covariates are to be included in the modeling process, then it is still possible to estimate cause-specific hazards but prediction of the cumulative incidences and survival probabilities based on these is no longer feasible. This article aims at providing some possible strategies for dealing with these prediction problems. In a multi-state framework, a first approach uses internal covariates to define additional (intermediate) transient states in the competing risks model. Another approach is to apply the landmark analysis as described by van Houwelingen [*Scandinavian Journal of Statistics* 2007, **34**, 70–85] in order to study cumulative incidences at different subintervals of the entire study period. The final strategy is to extend the competing risks model by considering all the possible combinations between internal covariate levels and cause-specific events as final states. In all of those proposals, it is possible to estimate the changes/differences of the cumulative risks associated with simple internal covariates. An illustrative example based on bone marrow transplant data is presented in order to compare the different methods.

Key words: Competing risks; Cumulative incidence function; Internal time-dependent covariates; Multi-state models; Semi-parametric regression models.

1 Introduction

Time-dependent covariates are frequently encountered in regression analysis of event history data. Information on individual variables measured during the study period, or information related to some features of the study being recorded or controlled over time, are often essential predictors which cannot be disregarded or substituted by time-fixed covariates. Sometimes, in some specific models for the hazards, time-dependent covariates, such as age that is defined as linear functions of time, do not contribute to explain time variation within individuals and can therefore be treated as time fixed.

Time-dependent covariates are also often used as tools in order to adjust for lack-of-fit, *e.g.* due to non-proportionality in a multiplicative hazard model, or to investigate other model assumptions such as the Markov property in multi-state models. In spite of the frequent use of time-dependent covariates, their role in regression modeling and prediction is, however, still unclear due to both interpretational and practical problems.

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This study briefly recalls the different types of time-dependent covariates, as classified by Kalbfleisch and Prentice (2002, Chapter 6), with the intent of clarifying their role and emphasizing the limitations in both the survival analysis and the competing risks setting. When different causes are acting simultaneously, the main interest is in estimating quantities such as cause-specific hazards, cumulative incidences or marginal survival probabilities. If random (*internal*) time-dependent covariates are to be included in the modeling process, then it is still possible to estimate cause-specific hazards but prediction of the cumulative incidences and survival probabilities based on these is no longer feasible (Andersen *et al.*, 1993, Chapter VII; Kalbfleisch and Prentice, 2002, Chapter 6). This limitation is also encountered in the regression approach for cumulative incidences (or "subdistribution hazards") (Fine and Gray, 1999) and has been further discussed by Latouche, Porcher, and Chevret (2005) and Latouche *et al.* (2007) and tackled by Beyersmann and Schumacher (2008).

This article aims at providing some possible strategies for dealing with these prediction problems. In a multi-state framework, this article presents some approaches for estimating cumulative incidences and survival probabilities when internal time-dependent covariates are included. The first approach uses internal time-dependent covariates to define additional (intermediate) transient states in the competing risks model. Another strategy is to apply the landmark analysis as described by van Houwelingen (2007) in order to study cumulative incidences at different subintervals of the entire study period. Here, the covariate is set to be time constant within each of those subintervals but allowed to vary between them. A final approach is to extend the competing risks model by considering all the possible combinations between internal covariate levels and cause-specific events as final states. In all of those proposals, it is possible to study the changes/differences of the cumulative risks associated with simple internal covariates.

We also show how, in an extended competing risks model, cumulative incidence probabilities may be allowed to depend on the sojourn times spent in transient states. Studying these sojourn times as time-dependent covariates may be useful both for testing model assumptions and for investigating their effect on the survival.

This article is structured as follows: Section 2 introduces briefly the different types of time-dependent covariates and explains the problems related to estimating cumulative probabilities, such as the survival function. In Section 3, the problem of accounting for the internal covariates following the subdistribution and the cause-specific hazard are discussed and compared. Section 4 describes the extended competing risks model and Section 5 discusses alternative ideas for including internal covariates into a competing risks framework when the interest is in estimating the cumulative incidence functions. Finally, an illustrative example based on real-life data is described in Section 6.

2 Time-Dependent Covariates

Given a specific study period $[0,\tau]$, a time-dependent covariate is a process $\mathbf{X}(t) = \{X(u); 0 \leq u \leq t\}$, $t \leq \tau$, which contains all the covariate information up to time t. Here, $X(t)$ is the value at time t. Under a given (independent) right-censoring scheme, we wish to specify a model for the hazard function $\alpha(t)$ for T, a life time variable, and let $\mathcal{F}_t^l$ be the filtration generated by the observed, right censored, survival data. In order to incorporate also information about covariates when a regression model is specified, an extended history F_t needs to be considered. If only time-independent covariates are present, this is $\mathcal{F}_t = \mathcal{F}_t^l \vee \mathcal{X}_0$, where $\mathcal{X}_0$ is the σ-algebra generated by the covariates observed at the time origin 0. In case of time-dependent covariates instead, the enlarged filtration is $\mathcal{F}_t = \mathcal{F}_t^l \vee \mathcal{X}_t$ and contains also covariate information up to time t, $\mathcal{X}_t$.

We assume that the covariate process $\mathbf{X}(t)$ is predictable with respect to $\mathscr{F}_t$, for instance that it has left-continuous paths (it is, by definition, already adapted). The conditional hazard function is then

$$\alpha(t;\mathbf{X}(t)) = \lim_{\Delta t \downarrow 0} \frac{1}{\Delta t} \mathrm{P}\{t \leq T < t + \Delta t | \mathbf{X}(t), T \geq t\}. \tag{1}$$

Time-dependent covariates can be divided into two general classes: *External* and *internal* covariates, sometimes also denoted as *exogenous* and *endogenous* covariates, respectively. An external covariate satisfies the condition

$$\mathrm{P}\{u \leq T < u + \Delta u | \mathbf{X}(u),\ T \geq u\} = \mathrm{P}\{u \leq T < u + \Delta u | \mathbf{X}(t),\ T \geq u\} \tag{2}$$

for all u and t such that $u \leq t$ (Kalbfleisch and Prentice, 2002, Chapter 6). Hence, the hazard function at time u is influenced by the observed covariate history up to time u by the regression model, but the occurrence of a failure in $[u,u+\mathrm{d}u)$ is independent of the future path of the covariate. For an internal covariate, condition (2) is not satisfied. The problem is that the possibility of observing the value of an internal covariate depends on the survival status of a subject. Therefore, observation of $X(\cdot)$ after u tells us that the subject will not fail at u and thus, Eq. (2) cannot hold. Typical examples of internal covariates are disease complications, measurements recorded at follow-up visits, such as biochemical and clinical characteristics, which may give prognostic information for patients. In general, external covariates have instead a path that is external to the individuals under study and not directly generated by their behavior in time. Some examples are the age of a patient, levels of air pollution or the time since disease diagnosis.

External covariates can vary in a predetermined way, that is their paths can be determined in advance (so-called *defined* covariates). Age of a patient and time since disease diagnosis (if time of diagnosis is included in the information available at the time origin) are examples of that. A time-fixed covariate also belongs to this class, since its (constant) value is given at the time origin. Certain random covariates such as measurements of air pollution can also be considered as external (so-called *ancillary* covariates), since their stochastic process has a distribution that does not involve the parameters of the regression model for survival times. Note that defined covariates are automatically adapted to the filtration $\mathscr{F}_t^{\prime} \vee \mathscr{X}$ (Andersen *et al.*, 1993), that is, their paths can be considered as being fixed in advance. Therefore, inference can be based on the partial likelihood conditional to the covariates, as usually done in case of time-constant covariates.

Ancillary and internal covariates, on the other hand, carry more randomness and their process $\mathbf{X}(t)$ is not adapted to the filtration $\mathscr{F}_t^{\prime}$. However, since the ancillary covariate processes are completely external to the individuals under study, the model for these covariates does not depend on the parameters of interest and does not need to be specified. Therefore, inference on the parameters of interest can still be based only on the partial likelihood, conditioning on the observed covariate paths. The main difference with an internal covariate is that it carries information about failure times of individuals. Therefore, we can write the hazard function only conditionally on the covariate process. $\mathbf{X}(t-)$ up to the time just before t. Inference can still be based on the partial likelihood when aiming at estimating hazard functions, although the full likelihood contains factors related to the conditional distribution of $\mathbf{X}(t)$ given the history $\mathscr{F}_t$ (Kalbfleisch and Prentice, 2002, Chapter 6). Thus, in this case we can avoid to specify a model for $\mathbf{X}(t)$. However, when the interest is on estimating the survival probability based on internal covariates, as well as the cumulative probabilities in more general contexts, inference can not only be based on the partial likelihood but would also need parameters from a joint model for T and the stopped process $X(t \wedge T)$, which then would need to be identified.

From these aspects, it follows that care needs to be taken in interpreting the survival function conditionally on internal covariates and the classical relation

$$\mathrm{P}(T > t) = \exp(-\int_0^t \alpha(u)\mathrm{d}u)$$

between survival function and hazard function no longer holds. The survival function is not anymore a function only of the hazard rate, but also of the random development of the covariates. Therefore, in order to provide an estimate for the survival probability, a model for the distribution of the stochastic process of the internal covariate also needs to be specified. For a categorical internal time-dependent covariate, such an extended, joint model for $\mathbf{X}(t)$ and T was discussed by Andersen (1986) and (Andersen Hansen, and Keiding 1991). In the next sections, that approach will be studied for the competing risks situation.

3 Competing Risks Regression Modeling and Internal Time-Dependent Covariates

First, we define the competing risks model and recall different approaches to regression analysis on time-fixed covariates. Next, we consider the problems encountered when including an internal time-dependent covariate $X(t)$.

The competing risks model is based on a stochastic process $\{Z(t), t \in [0, \tau]\}$ in continuous time, with right-continuous paths and a finite state space, $\mathscr{S} = \{0, 2, \ldots, \mathrm{k}\}$. Note that we skip index 1, since it is appropriately used later on in the manuscript. Let $\mathscr{F}'_t$ be the σ-algebra generated by the process $Z(t)$. The only transient state is 0, whereas the remaining states $\{2, \ldots, k\}$ are absorbing. Usually, as often in an epidemiological context, the absorbing states represent different causes to which the event under study may be due. Let T be the time until the event occurs. Each of the transitions from the state 0 to the states $\{2, \ldots, k\}$ represents the occurrence of the event due to cause h, with $h \in \{2, \ldots, k\}$ (where the same notation is used for the absorbing states and for the causes). Given a realization of the process $Z(t)$, at most one of these transitions is observed. The distribution of a competing risks process is regulated by the transition probability matrix

$$\boldsymbol{P}(s, t) = \{P_{lh}(s, t), l, h \in \mathscr{S}\},$$

where the non-trivial transition probabilities are $P_{0h}(s, t) = \mathrm{P}(Z(t) = h | Z(s) = 0)$, with $h \in S$ and $s < t$. Under a competing risks setting, for $h \in \{2, \ldots, k\}$, these probabilities are commonly denoted as cumulative incidence functions. One important aspect of the model is that it belongs to the class of non-homogeneous Markov processes, with the following Markov property

$$P_{lh}(s, t) = \mathrm{P}(Z(t) = h | Z(s) = l, \mathscr{F}'_{s-}) = \mathrm{P}(Z(t) = h | Z(s) = l). \tag{3}$$

The process $Z(t)$ may, equivalently, be determined by specifying all the positive transition intensities related to $\boldsymbol{P}(s, t)$,

$$\alpha_h(t) = \lim_{\Delta t \to 0} \frac{\mathrm{P}(t \le T < t + \Delta t, Z(T) = h | T \ge t)}{\Delta t}, \quad h = 2, \ldots, k. \tag{4}$$

Note that, under a competing risks model, we write $\alpha_{lh}(t) = \alpha_h(t)$ since the only positive transition intensities are for $l = 0$. The $\alpha_{\mathrm{h}}(t)$ are known as the cause-specific hazard functions and they are depicted in panel (A) of Fig. 1.

3.1 Regression models for cause-specific hazards

To study how covariates affect the transition probabilities in a competing risks model, a regression analysis can be performed by following different approaches. The simplest, classical, approach (Prentice *et al.*, 1978; Andersen Abilstrom, and Rosthøj 2002) is to build a standard regression model for each cause-specific hazard. These separate models may then, subsequently, be combined to a prediction of the transition probabilities in $\boldsymbol{P}(s,t)$ using the relations between the two sets of functions. These are

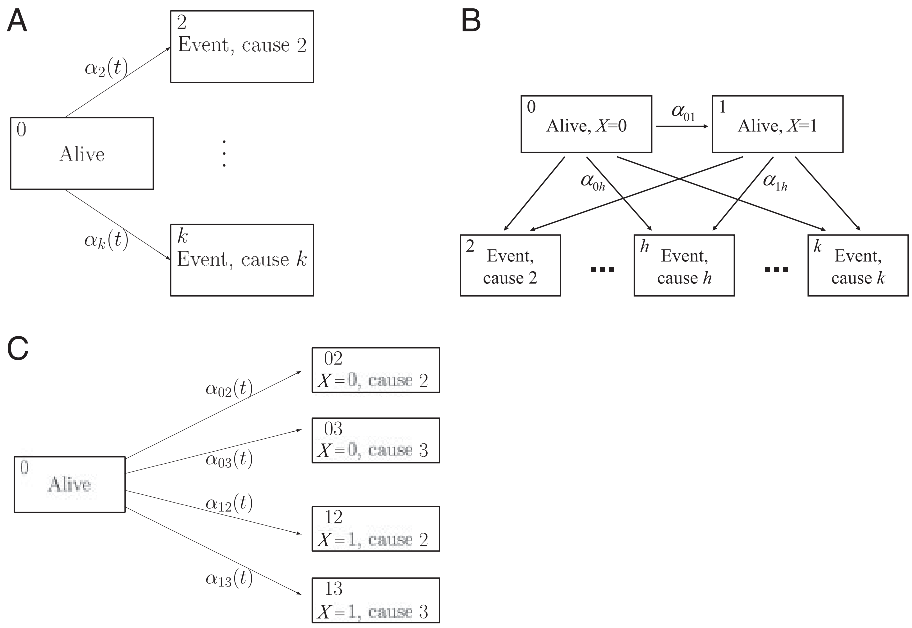

Figure 1 (A): Competing risks model with k-1 different causes of the event under study. (B): The extended competing risks model with a transient state, without possibility of recovery. The ending event can be due to causes $\{2,\ldots,k\}$. The arrows represent the possible transitions between states. (C): Modified competing risks model with four final states (two covariate levels and two causes) for including information about a categorical internal time-dependent covariate.

$$S(t) = P_{00}(0,t) = \exp\left(-\sum_{h=2}^{k}\int_0^t \alpha_h(u)\mathrm{d}u\right)$$

and

$$P_{0h}(0,t) = \int_0^t S(u-)\alpha_h(u)\mathrm{d}u, \quad h=2,\ldots,k.$$

Regression models for two cause-specific hazards can have different and common covariates, as well as different and common parameters. When analyzing the regression model for the cause-specific hazard due to cause h, failures due to the competing causes are, technically, treated as censored observations because the likelihood factorizes.

These equations also hold when the models for the cause-specific hazards include an external time-dependent covariate $X(t)$ (Kalbfleisch and Prentice, 2002, Chapter 8):

$$S(t;X(t)) = P_{00}(0,t;X(t)) = \exp\left(-\sum_{h=2}^{k}\int_0^t \alpha_h(u;X(u))\mathrm{d}u\right) \quad (5)$$

and

$$P_{0h}(0,t;X(t)) = \int_0^t S(u-;X(u-))\alpha_h(u;X(u))\mathrm{d}u, \quad h=2,\ldots,k. \quad (6)$$

3.2 The Fine–Gray model

A regression model for the cause-specific hazards does not provide a simple regression parameter that measures the direct effect of covariates on the cumulative incidence function, and separate regression analyses of all the cause-specific hazards need to be performed even when the interest is only on a specific cause and the related probability. These difficulties have led to alternative approaches that aim to establish a direct effect of covariates on the cumulative incidence probabilities through a regression model. In the following, we will focus on the approach for competing risks regression suggested by Fine and Gray (1999) and its limitation when time-dependent covariates are studied.

The Fine–Gray model introduces covariates in a model for the cumulative incidence *via* the so-called "subdistribution hazard function" originally introduced by Gray (1988). We assume that the cause of interest is cause 2 and the subdistribution hazard, $\alpha_2^*(t; X)$, for that cause is then defined by the relationship

$$P_{02}(0, t; X) = 1 - \exp\left(-\int_0^t \alpha_2^*(u; X)\mathrm{d}u\right), \tag{7}$$

where for the moment we assume that covariates are time independent. The function $\alpha_2^*(t; X)$ has the following interpretation

$$\alpha_2^*(t; X) = \lim_{\Delta t \to 0} \frac{\mathrm{P}(t \leq T < t + \Delta t, Z(T) = 2 | (T \geq t) \cup (T \leq t, Z(T) \neq 2), X)}{\Delta t}. \tag{8}$$

The $\alpha_2^*(t; X)$ corresponds to the hazard function of the modified survival time $T^* = T \cdot I(Z(T) = 2) + \infty \cdot I(Z(T) \neq 2)$, which has an improper distribution function equal to $P_{02}(0, t; X)$ for $t < \infty$ and a point mass $\mathrm{P}(T < \infty, Z(T) \neq 2; X)$ at $t = \infty$. This means that individuals failing from causes different than 2 remain at risk for all $t < \infty$, although they can no longer experience a failure from cause 2.

Though the subdistribution hazard, by conditioning on possible failure from other causes before time t, has an awkward interpretation, the Fine–Gray model is, anyhow, useful by establishing a direct relationship between the cumulative incidence of interest and the covariates. When fitting the model to right-censored data some complications arise. This is because for those subjects not failing from cause 2 of interest, the minimum of T^* and the censoring time C cannot be observed. Therefore, one needs to consider, in addition, the censoring distribution in order to estimate the probability of being censored over time for each individual, thereby applying inverse probability of censoring weighting methods (Robins and Rotnitzky, 1992).

3.3 Internal covariates

Consider now an internal time-dependent covariate $X(t)$. Then, it is still possible to include $X(t)$ in a model for the cause-specific hazards, as in standard hazard regression models like the proportional hazards model (Cox, 1972). Therefore, ratios between cause-specific hazards associated with $X(t)$ can be estimated. However, the right-hand sides of Eqs. (5) and (6) cannot anymore be interpreted as probabilities. The reason for that was already explained at the end of Section 2 for the survival probability. Similarly here, $S(t; X(t)) = \mathrm{P}\{Z(t) = 0 | X(t)\} = \mathrm{P}\{T > t | \mathbf{X}\}$ in Eq. (5), with $X(t)$ being an internal covariate, would be equal to one, since the observation of X at time t yields information on the survival status at that time. Prediction of $S(t; X(t))$ and $P_{0h}(0, t; X(t))$, which depends on the survival probability as in Eq. (6), is thus not possible, given an internal time-dependent covariate.

Similarly, in the subdistribution hazard approach the equality in Eq. (7) does no longer hold, since it is not possible to write the cumulative incidence probability $P_{02}(0, t; X(t))$ as a function of the hazards when the model includes internal covariates. The focal point of this problem is the same as already explained in Section 2.

Moreover, in the subdistribution hazard approach, an additional interpretational problem concerns all types of time-dependent variables, and it appears conceptually similar to the one explained for the censorship. We again assume that 2 is the cause of interest. The question is, for those individuals who failed from causes different than 2, how $X(t)$ should be defined on the interval between their real failure time T and their, possibly unobserved, potential censoring time C, if the time-dependent covariate is no longer observed on this interval. The proposal by Beyersmann and Schumacher (2008) provides a solution for handling the interpretational problem of general time-dependent covariates, and thus overcomes the limits given by the regression of the subdistribution hazard on an internal covariate. Their solution amounts to studying the stopped covariate process $X(t \wedge T)$, which is well defined even after the time, T, of failure. Therefore, under a proportional hazards specification, the subdistribution hazards ratio associated with $X(t)$ can be estimated. As an example, given the model $\alpha_2^*(t;X) = \alpha_{2,0}^*(t)\exp(bX(t))$ with $\alpha_{2,0}^*(t)$ being the baseline hazard, the regression coefficient b can be estimated in order to explain the effect of an internal covariate $X(t)$ on the subdistribution hazard. Because of the non-direct interpretation of $\alpha_2^*(t;X)$, this feature is of minor use. The most useful property of a subdistribution hazard is its one-to-one relationship with the cumulative incidence but, unfortunately, the Beyersmann and Schumacher method does not provide a regression model for the cumulative incidence function.

In this article, we will argue that the influence of simple internal time-dependent covariates on $P_{02}(0,t;X(t))$ may be studied using a multi-state approach, though it is not possible to estimate a single regression parameter for the direct covariate effects on the cumulative incidence of interest. In particular, the competing risks multi-state model in panel (a) of Fig. 1 may be extended in different ways in order to allow for categorical internal covariates. In these extensions, comparisons between estimated cumulative probabilities can reveal the impact of those covariates on the cause-specific risks. The following sections present two extended competing risks models and a third idea for dealing with internal covariates.

4 An Extended Competing Risks Regression Model with Transient States

When the internal time-dependent covariate studied in a competing risks setting is categorical, we may incorporate the information carried by the covariate process into the competing risks model. A possible way of doing that is to represent each possible value that the covariate can assume over time as a transient state, as also mentioned by Beyersmann and Schumacher (2008) and Andersen (1986).

For sake of simplicity, we study a simple binary time-dependent covariate $X(t) \in \{0,1\}$, and we define $T_1 = \inf\{t \in [0,\tau] : X(t) = 1\}$. Referring to the same notation used at the beginning of Section 3, the competing risks model is extended by adding a transient state which corresponds to the covariate value $X(t) = 1$ while state 0 now corresponds to $X(t) = 0$; T_1 is thus the waiting time to transition into state 1. Panel (B) of Fig. 1 shows this model with all possible transitions regulated by the positive hazard functions $\alpha_{0l}(\cdot)$, for $l \in \{1,\ldots,k\}$, and $\alpha_{1h}(\cdot)$, for $h \in \{2,\ldots,k\}$. Note that, by writing the intensities out of state 1 as depending only on t (and not on T_1), we have assumed that the extended multi-state process is still Markovian. We will later relax this assumption. In summary, the state space $\mathscr{S}$ is enlarged to $\mathscr{S}' = \{,,,\ldots,\mathscr{k}\}$ and the multi-state process $Z(t)$ is modified so that $Z'(t) = X(t+)$ in the open time interval $[0,T)$.

The transition probabilities in the extended model are now given, as follows. The probability

$$P_{01}(s,t) = \int_s^t P_{00}(s,u-)\alpha_{01}(u)P_{11}(u,t)\mathrm{d}u, \quad s \le t, \quad s,t \in [0,\tau] \tag{9}$$

depends on the probabilities of permanence in states 0 or 1 in subsets of the time interval $[s,t]$. These have the explicit form

$$P_{hh}(s,t) = \exp\left\{\int_s^t -\alpha_{hh}(u)\mathrm{d}u\right\}, \quad h = 0,1, \quad s \le t, \tag{10}$$

where

$$\alpha_{00}(u) = \alpha_{01}(u) + \sum_{h=2,\ldots,k} \alpha_{0h}(u), \qquad \alpha_{11}(u) = \sum_{h=2,\ldots,k} \alpha_{1h}(u).$$

The probability $P_{01}(s,t)$ has the following intuitive interpretation: an individual may stay in state 0 for a certain time $[s,u\text{-}]$ with probability $P_{00}(s,u\text{-})$, then he may jump to state 1 at u with instantaneous rate $\alpha_{01}(u)$ and finally stay in state 1 the remaining time $[u,t]$ with probability $P_{11}(u,t)$. The transitions from state 1 to the final states are regulated by the probabilities

$$P_{1h}(s,t) = \int_s^t P_{11}(s,u-)\alpha_{1h}(u)\mathrm{d}u, \quad h = 2,\ldots,k, \quad s \leq t. \tag{11}$$

The cumulative incidence probabilities for the final events are the transition probabilities from state 0 to the absorbing states $2,\ldots,k$, given by

$$P_{0h}(s,t) = \int_s^t [P_{00}(s,u-)\alpha_{0h}(u) + P_{01}(s,u-)\alpha_{1h}(u)]\mathrm{d}u, \quad h = 2,\ldots,k, \quad s \leq t. \tag{12}$$

The marginal survival probability, that is the probability of not experiencing any event until time t, is given by

$$S(t) = P_{00}(t) + P_{01}(0,t). \tag{13}$$

For non-homogeneous finite-state Markov processes, the quantities illustrated above, and thus the transition matrix $\boldsymbol{P}$, can be estimated by the Aalen–Johansen estimator (Aalen and Johansen, 1978). Estimation of $P_{0h}(\cdot)$ in Eq. (12), $P_{1h}(\cdot)$ in Eq. (11) and $S(\cdot)$ in Eq. (13) is straightforward, since it is obtained by plugging in estimates $\hat{P}_{00}$ and $\hat{P}_{11}$ of P_{hh} given in Eq. (10).

An example of such a context could be when, in a clinical study, it is important to take the binary covariate response/no response to a certain treatment into account. The extended competing risks model would result in an initial state 0 and an intermediate state 1, corresponding to, respectively, "no response" and "response." Therefore, the probabilities of dying from a certain cause, for response and non-response patients, can be estimated and even compared. It may also be of interest for instance to estimate the probability $P_{01}(0,t)$ of being in response at a certain time t.

The extended competing risks model would also allow patients to enter the study in either state 0 or 1, accounting then for left truncation. An initial distribution for entering in either state 0 or state 1 could also be specified. For example, when studying mortality due to different causes for certain children, taking the role of HIV into account, one might be faced with children with or without HIV at birth, corresponding to entering in state 1 or 0, respectively.

When the binary internal covariate is defined as a more general process which can also jump backward to the value 0, it is still possible to extend the competing risks model as described above, but the possibility of "recovery" needs also to be allowed, that is, the transition from state 1 back to state 0 is possible. In this case, explicit expressions for the transitions probabilities can no more be obtained, although estimators based on the product integral representation can be still computed (Andersen *et al.*, 1993, Chapter IV).

4.1 Sojourn time in the intermediate states

In the extended competing risks model described above, one would need to be able to check the Markovian assumption behind this model. A possible method is to investigate if the sojourn time into state 1 influences the possible following transitions to the final states. Suppose we are interested in studying the duration at time t in state 1 since the time, T_1, of entry into this state. Denote this duration by $d = (t - T_1)^+$. We can allow the intensities regulating the transitions from 1 to the final states to depend on d, that is, $\alpha_{1h}(t,d)$, $h = 2,\ldots,k$. The resulting model is then a semi-Markov model.

The intensities $\alpha_{1h}(t,d)$ may be modeled by regression on the time-dependent covariate d, besides other covariates. An example of that is the Cox regression model, $\alpha_{1h}(t,t-T_1) = \alpha_{1h,0}(t)\exp\{(t-T_1)\beta\}$. Since the transition time T_1 is random, the duration $d=(t-T_1)^+$ is a simple random process, which is null until T_1, while after T_1 it increases linearly with time t. Let $\mathcal{F}_t^I$ and $\mathcal{X}_t$ be the filtrations generated by the observed multi-state process and the observed covariate d, respectively. Since T_1 is determined by the multi-state process itself, d is adapted to the filtration $\mathcal{F}_t^I$ generated by the observed multi-state process, and then the observed history of d, $\mathcal{X}_t$, is so that $\mathcal{X}_t \subset \mathcal{F}_t^I$ and thereby $\mathcal{F}_t = \mathcal{F}_t^I \vee \mathcal{X}_t = \mathcal{F}_t^I$. The covariate d cannot be considered as determined in advance from time $t=0$, since T_1 is unobserved at the time origin. However, after entering state 1 at T_1, given the covariate history $\mathcal{X}_t$ with $t \geq T_1$, the duration d can be considered as determined. Thus, it may be thought of as a *defined* time-dependent covariate when $P_{1h}(s,t)$ are modeled.

Since d is assumed to influence the multi-state process only through the transition intensities $\alpha_{1h}(\cdot), h=2,\ldots,k$, the transition probabilities from state 1 given the observed history $\mathcal{F}_{s-}$ are

$$P_{1h}(s,t|T_1) = \int_s^t P_{11}(s,u-|T_1)\alpha_{1h}(u,u-T_1)\mathrm{d}u, \quad h=2,\ldots,k, \quad T_1 < s \leq t,$$

where

$$P_{11}(s,u|T_1) = \exp\left(-\int_s^u \sum_{h=2}^{k} \alpha_{1h}(v,v-T_1)\mathrm{d}v\right).$$

Similarly to Eq. (12), the cumulative incidence probabilities for $h=2,\ldots,k$ can be expressed as follows

$$\begin{aligned} P_{0h}(s,t) &= \int_s^t [P_{00}(s,u-)\alpha_{0h}(u) + P_{01}(s,u-)\alpha_{1h}(u)]\mathrm{d}u \\ &= \int_s^t P_{00}(s,u-)\alpha_{0h}(u)\mathrm{d}u + \int_s^t P_{00}(s,u-)\alpha_{01}(u)P_{1h}(u,t|u)\mathrm{d}u, \end{aligned} \tag{14}$$

where

$$P_{1h}(u,t|u) = \int_u^t P_{11}(u,v-|u)\alpha_{1h}(v,v-u)\mathrm{d}v,$$

and P_{00} is given by (10).

When the transition intensities depend on $d=(t-T_1)^+$, which is studied as a time-dependent covariate, estimation of the transition probabilities is also straightforward, since in Eq. (14) the conditional probabilities $P_{1h}(u,t|u)$ are computed for all possible times u of transition into state 1. Some complications may arise when left truncation is present, since information on T_1 may not be known for patients with delayed entry. If individuals are allowed to enter the study in either state 0 or 1, in general predictions on P can be possible only if the filtration $\mathcal{F}_0$ at time 0 contains information about the previous entry time T_1 in state 1.

5 Alternative Approaches for Competing Risks Regression with Categorical Internal Covariates

Let us now, again for sake of simplicity, restrict attention to the case of two competing causes. Further, we suppose that the regression analysis needs to involve a simple binary internal covariate $X(t) \in \{0,1\}$, defined as a one-step process $\mathbf{X}(t) = I(T_1 > t)$ with $T_1 = \inf\{t \in [0,\tau] : X(t) = 1\}$. This covariate may be equal to 0 at the time origin for all individuals, and it can possibly change its value to 1 at a certain time, T_1, and, in that case, it remains time-constant and equal to 1 over the remaining time. If $T_1 = 0$, the covariate may also take the value 1 for all $t > 0$.

The extended multi-state model considered in Section 4 required a model for the development of the internal time-dependent covariate over time. Although the initial Markov assumption may be tested and relaxed, the validity of the inference will, to some extent, depend on the assumptions imposed in the model for $X(t)$. In this section, we will introduce two simple ways of examining how $X(t)$ affects the cumulative incidences without having to specify a model for $X(t)$. Both methods are based on predicting at a number of "landmark time points."

One possible strategy for investigating the dependence of such a time-dependent covariate on the cumulative failure risks is to enlarge the set of final states $\{2, 3\}$ of the original competing risks model, so that the information of the covariate status at the failure time is also taken into account. The modified model, illustrated in panel (C) of Fig. 1, has four possible final states that we denote as 02, 03, 12 and 13. The individuals who fail at time T from cause 2 or 3 and have covariate value $X(T) = 0$, that is the covariate status at the failure time is zero, experience a transition to state 02 or 03. On the other hand, individuals failing from cause 2 or 3, but with covariate value $X(T) = 1$ at that time, have a transition to state 12 or 13.

Under this model, subjects are allowed to have any covariate value $X(t)$ for $t \in [0, \tau]$, but of course those with $X(t) = 1$ cannot experience transitions to $\{02, 03\}$. The idea is now to focus on the following conditional cumulative incidences for suitable values of "landmarks" s:

$$P_{0h}(s, t; X(s)) = \mathrm{P}(T \leq t, Z(T) = h \big| T \geq s, X(s)), \quad h \in \{02, 03, 12, 13\}. \tag{15}$$

It is important to note that the quantity in Eq. (15) is conditioned only on a time-constant covariate, since $X(s)$ contains only the information of the covariate status at the initial time point s, where it is known that the subject is still alive. Thus, the conditional probabilities in Eq. (15) contain only the covariate information up to time s, ignoring the future covariate development. A subject with $X(s) = 0$ can still change its covariate status to $X(T_1) = 1$ at a certain time T_1 between s and t, and this information is then recorded by the final state where the subject possibly ends. On the other hand, an individual with $X(s) = 1$ can no more change his future covariate status, therefore no additional covariate information over the future time is needed, and he is constrained to possibly end in state 12 or 13.

By estimating the cumulative probabilities in Eq. (15) at different time points s, for $0 < s < t$, conditional on the different covariate values, it is possible to describe the effect of $X(t)$ on the cause-specific failure risks over time. Therefore, one can see how this effect modifies the estimated cumulative incidence. The estimation procedure is straightforward and can be based on the product integral estimator (Aalen and Johansen, 1978; Andersen *et al*., 1993, Chapter VII).

Our second suggestion is to directly adopt "landmark analysis" as suggested by van Houwelingen (2007). Consider a simple competing risks model, as defined in Section 3. This procedure consists of a series of simple (competing risks) survival analyses with time-fixed covariates conducted at a number of landmark time points, s. More specifically, one would estimate

$$\mathrm{P}(T \leq t, Z(T) = 2 | T \geq s, X(s)), \tag{16}$$

that is the cumulative incidence given the status at s, by looking only at those subjects alive at s. This can be done for $s = 0$, as also shown by Beyersmann and Schumacher (2008) in their example, but also for later values of s.

Computing these probabilities for different landmarks s requires to use the restricted samples of patients still alive at each s. For $s = 0$, the probability in Eq. (16) is just the usual cumulative incidence given $X(0)$, while, for later values of s, we have conditional cumulative incidences given survival until s and given $X(s)$. Other covariates may be adjusted for in the analysis using, *e.g*. the Fine–Gray model.

Note that $X(s)$ is a time-constant covariate when Eq. (16) is estimated at each s. In fact, for a given landmark time s, it is only the covariate value at s, $X(s) = 1$ or $X(s) = 0$, that is accounted for, while future values of $X(u), u > s$, are not taken into account. However, the covariate $X(s)$ is allowed to vary between the sequence of landmarks s. In this way, by setting a number of landmarks s at opportune time points, the sequence of probabilities $P(T \leq t, Z(T) = 2 | T \geq s, X(s) = 0)$ and $P(T \leq t, Z(T) = 2 | T \geq s, X(s) = 1)$ may be compared, and hence giving insight into how $X(\cdot)$ affects the failure risks.

6 Example on Bone Marrow Transplant Studies

The worked example is based on data from bone marrow transplant patients collected by the Center for International Blood and Marrow Transplant Research (CIBMTR). Our data set concerns patients who received an HLA-identical sibling transplant from 1995 to 2004 for acute myelogenous leukemia (AML) or acute lymphoblastic leukemia (ALL) and were transplanted in first complete remission. All patients received bone marrow transplantation or peripheral blood stem cell transplantation. The infants aged less than 2 years old and all patients who received umbilical cord blood transplants were excluded, as risk factors are likely to vary in this group.

The data consist of 2009 patients with mean age equal to 31.9 years (SD = 15.4). Most of the patients (70%) underwent transplant for AML disease. The time variable considered in our analyses was time (in months) since transplant. During this period, 976 (49%) patients developed Graft *versus* Host Disease (GvHD); there were 259 cases (13%) of relapse and, among them, 232 (90%) patients died. The total number of deaths was 737 (37%). For ease of presentation, we have combined information on acute and chronic GvHD into one time-dependent variable.

In all the following regression analyses, age (centered by subtracting its mean) and disease (AML *versus* ALL) were the only time-fixed covariates we included in the models. This choice was made in order to keep the explanation of the different estimated transition probabilities simpler. We focused on studying the influence of the internal time-dependent variable GvHD, by following the different approaches described in this article. The analyses were carried out with the survival package (version 2.35-4, 2009) of the R software, version 2.9.0 (R Development Core Team, 2009, `http://www.R-project.org`). The computer code used to produce the current example is available online at the webpage `http://homes.stat.unipd.it/gcortese`.

The example is structured as follows. We first consider a standard three-state-competing risks model. Within this model, cumulative cause-specific hazards and cumulative incidence functions were estimated both non-parametrically without regarding covariates and using regression on the two time-fixed variables. In a subsequent analysis, we studied the case when the time-dependent covariate GvHD is included in the cause-specific hazard regression models, and the cumulative cause-specific hazards are estimated. Successively, we studied the extended models explained in Sections 4 and 5, both without and with including time-fixed covariates and thereby providing estimates of the (conditional) cumulative incidence functions. Finally, within the standard three-state competing risks model, we applied the "landmark" approach for estimating the cumulative incidence functions conditioning on the status at different time points.

6.1 The three-state competing risks model

The three states of the competing risks model are defined as the initial state 0: alive in remission after transplant and the two absorbing states 2: relapse and 3: death. In the competing risks model with no covariates, cumulative incidence functions were estimated using the Aalen–Johansen estimator and cumulative cause-specific hazards were based on the Nelson–Aalen estimator. The estimated curves in Fig. 2 show a high risk of death compared with the risk of relapse.

For the competing risks model with only time-fixed covariates, Cox regression models on the cause-specific hazards were assumed for the relapse and mortality rates. The corresponding estimates are reported in the left-hand side of Table 1. They suggest that age significantly affects only the death rate, increasing it by a hazard ratio of HR = 1.03 per year, (CI = [1.02, 1.04]). The AML disease, compared with ALL, is associated with both a higher relapse rate (HR = 1.73, CI = [1.35, 2.23]) and a higher death rate (HR = 1.50, CI = [1.23, 1.81]). The predicted cumulative incidence probabilities (data not shown here) have a very similar behavior to those shown in Fig. 2 for the case of no covariates. They were found to increase for older patients and for AML disease for both relapse and death, as expected by looking at the estimated cause-specific hazard rates in the two Cox models.

If the time-dependent covariate GvHD is also included in the two regression models for the death and relapse rates, estimates related to age and disease remained almost unchanged, as seen in the right-hand side of Table 1. GvHD seems not to be an important risk factor for the rate of relapse, although a trend toward a protective effect is suggested, while appearance of GvHD during follow-up increased the mortality rate significantly (HR = 2.85, CI = [2.36, 3.44]). Under this model, it makes sense only to estimate the cumulative hazard functions for relapse and death, as shown in Fig. 3 for a 44-year-old patient with AML disease, for different times of appearance of GvHD. We observe that the cumulative hazard for relapse decreases slightly when GvHD appears earlier in time, while the effect of GvHD on the cumulative hazard for death is seen clearly by the pronounced difference between lines in the right-hand panel of Fig. 3.

6.2 An extended competing risks model with an intermediate state

We studied the extended competing risks model with GvHD as an intermediate transient state (denoted here by 1). In this model, transitions from state GvHD back to the initial state 0 are not possible. A patient may experience directly relapse or death with no event of GvHD, or he or she can first develop GvHD, moving thus to state 1, and then experience one of the final events relapse or death.

First, the transition probabilities between states were estimated non-parametrically, as shown in Fig. 4 (panels (a) and (b)). Then, separate Cox regression models with the two time-fixed covariates age and disease were assumed for each transition intensity. Under this extended competing risks

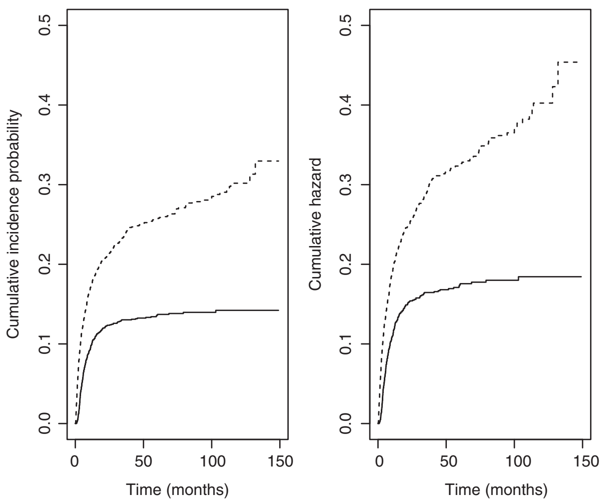

Figure 2 Cumulative incidence functions, estimated by the Aalen–Johansen estimator (left-hand side), and cumulative cause-specific hazards, based on the Nelson–Aalen estimator (right-hand side), in the three-state competing risks model. The solid lines are for the event of relapse and the dashed lines are for the event of death.

Table 1 Estimates from the Cox model for the relapse and mortality rates. The left-hand side of the table refers to models where only time-fixed covariates are included, the right-hand side to models with the additional time-dependent covariate GvHD.

	Cox model for relapse (cause 2)					
	Without GvHD			With GvHD		
Covariates	β	SE(β)	*p*-Value	β	SE(β)	*p*-Value
Centered age	−0.003	0.004	0.430	−0.003	0.004	0.560
AML disease	0.550	0.129	<0.001	0.563	0.129	<0.001
GvHD	–	–	–	−0.172	0.133	0.190
	Cox model for death (cause 3)					
	Without GvHD			With GvHD		
Covariates	β	SE(β)	*p*-Value	β	SE(β)	*p*-Value
Centered age	0.030	0.003	<0.001	0.027	0.003	<0.001
AML disease	0.400	0.097	<0.001	0.332	0.098	<0.001
GvHD	–	–	–	1.048	0.097	<0.001

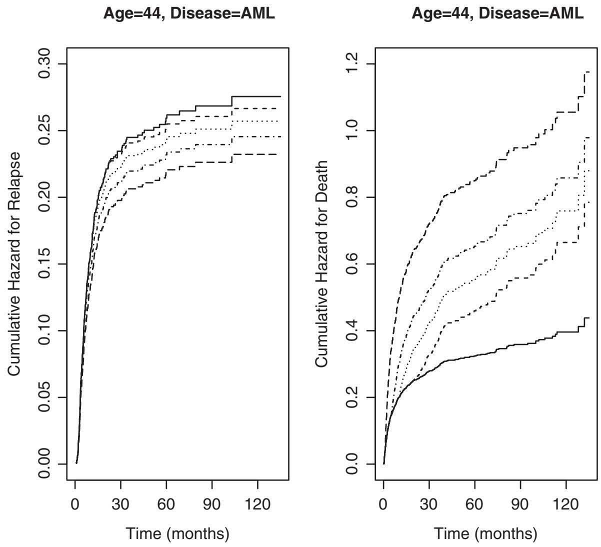

Figure 3 Three-state competing risks model (Section 3): Estimated cumulative hazard functions for relapse (left) and death (right), under a Cox model with age, disease and GvHD for the relapse and mortality rates. The curves represent an individual with AML disease and age 44 years. Solid lines represent no appearance of GvHD over follow-up time, while presence of GvHD developed at times $t = 20,10,5,1$ corresponds, respectively, to the dashed, dotted, dotted–dashed and long dashed lines.

model, estimates of the regression coefficients for the cause-specific rates α_{02}, α_{03} were very similar to those for the rates α_{12}, α_{13} (see Table 2), and also they did not differ much from the estimates from the previous three-state competing risks model. In both the cases with and without covariates, the estimated cumulative incidence probabilities for relapse, as well as the estimated cumulative hazards for relapse (solid lines in Fig. 4), are slightly lower (though very similar) for patients who developed GvHD over time, compared with no GvHD patients. If we look at the estimated transitions to death (dashed lines in Fig. 4), we see that this extended competing risks model is able to capture well how the GvHD status affects the cumulative incidence. Patients who experienced GvHD, going thus to state 1 of the model, had a considerably higher death risk than patients who did not develop GvHD. This fact was also suggested by comparing the cumulative hazards for death in Fig. 3.

6.3 An extended competing risks model with additional final states and the "landmark analysis"

Starting from our simple time-dependent internal covariate $X(\cdot)$, representing GvHD, with values in $\{0,1\}$, we built a competing risks model with the enlarged state space $\mathscr{S}=\{,,,,\}$, as described in Section 5 and panel (C) of Fig. 1. We studied such a model with no covariates, and focused on the differences between $P_{0,02}(s,t|X(s)=0)$ and $P_{0,12}(s,t|X(s)=1)$ for relapse, and between $P_{0,03}(s,t|X(s)=0)$ and $P_{0,13}(s,t|X(s)=1)$ for death, observed at various successive landmark times s. On the subset of patients with $X(s)=0$ at time s, we obtained estimates of the probabilities $P_{0,h}(s,t|X(s)=0)$ for $h\in\{02,03,12,13\}$, while we used the subset of patients with $X(s)=1$ at time s to estimate the probabilities $P_{0,h}(s,t|X(s)=1)$ for $h=12$ and $h=13$ only, since those latter are null for the other final states.

Figure 5 refers to the event of relapse. At time $s=0$ (top left panel) the estimated curves for $P_{0,02}(s,t|X(s)=0)$ and $P_{0,12}(s,t|X(s)=0)$ do not differ substantially, and they indicate a slightly lower cumulative risk of relapse in presence of GvHD, compared with patients without GvHD. This conclusion is very similar to what was already observed by comparing the solid lines in Fig. 4 (panel (b)) under the extended competing risks model. The difference between the same two probabilities

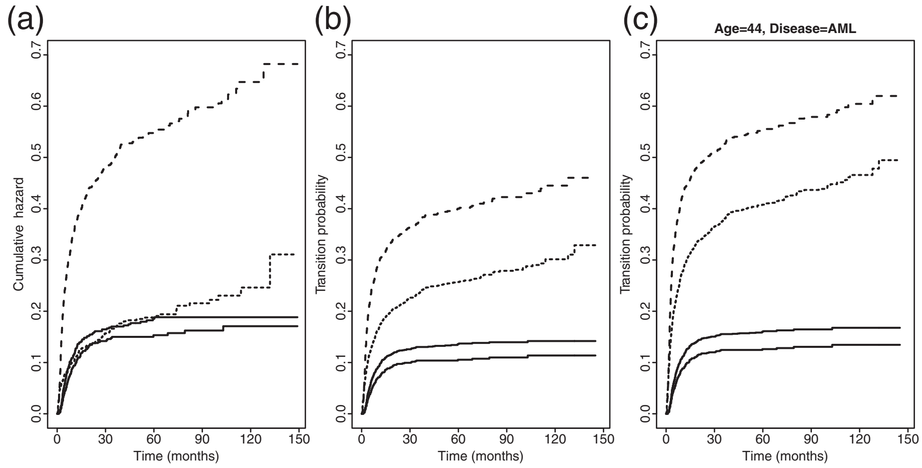

Figure 4 Extended competing risks model (Section 4): Estimates of cumulative cause-specific hazards (panel (a)) and incidence functions (panels (b) and (c)). The case of no covariates is represented in panels (a) and (b), the case with the time-fixed covariates age and disease is shown in panel (c). Solid lines are for the event of relapse (cause 2), thin and thick solid lines refer to the transitions $0\rightarrow2$ and $1\rightarrow2$, respectively (state 0 means "alive with no GvHD," state 1 means "alive with GvHD"). Dashed lines are for the event of death (cause 3), thin and thick dashed lines refer to the transitions $0\rightarrow3$ and $1\rightarrow3$, respectively.

Table 2 Estimates from Cox regression models for each transition rate in the extended competing risks model (Section 4).

	$0\to2$			$0\to3$		
Covariates	β	SE(β)	p-Value	β	SE(β)	p-Value
Centered age	−0.006	0.005	0.270	0.026	0.005	<0.001
AML disease	0.564	0.160	<0.001	0.267	0.159	0.093
	$1\to2$			$1\to3$		
Covariates	β	SE(β)	p-Value	β	SE(β)	p-Value
Centered age	0.002	0.007	0.720	0.029	0.004	<0.001
AML disease	0.585	0.125	0.008	0.361	0.125	0.004
	$0\to1$					
Covariates	β	SE(β)	p-Value			
Centered age	0.016	0.002	<0.001			
AML disease	0.289	0.070	<0.001			

($P_{0,02}(s,t|X(s)=0)$ and $P_{0,12}(s,t|X(s)=0)$) is clearly more pronounced at the following landmark times, since GvHD occurred mostly within the first month since transplant. Therefore, the probability of getting relapse with GvHD, given that GvHD did not occur until time s, becomes lower very fast for later times s. The comparison of the thick solid lines ($P_{0,12}(s,t|X(s)=1)$) with the dashed lines ($P_{0,02}(s,t|X(s)=0)$) in Fig. 5 shows that they are overlapping for all landmark times. Thus, we may conclude that individuals with GvHD and those who had not developed GvHD at time s, have the same probability for occurrence of relapse during time after s. In these two patient groups, the probability for death is, instead, much increased by the presence of GvHD, as it can be concluded by observing the very high thick solid lines running above the others in Fig. 6.

Finally, we applied the "landmark analysis" suggested by van Houwelingen (2007) to the bone marrow transplant data. We considered a simple competing risks model with the two final states relapse (cause 2) and death (cause 3). The analysis needed to set a series of landmark times s, which were chosen here as $s = 1, 2, 4, 6, 8, 10, 20$, as also done for the extended competing risks model in the current subsection. At each of these s, a competing risks regression analysis was performed only on the patients still alive at s and the covariate GvHD at time s was included as time fixed in the Cox models for the cause-specific hazards. A Cox regression model was fitted separately for the relapse rate and the death rate, at each s. Therefore, with this procedure, the regression coefficients have been estimated again for each landmark s, and also different estimates were obtained for the cause-specific baseline hazards.

Predictions of the conditional cumulative incidence functions $P_{0h}(s,t|T\geq s, X(s))$ were compared for $X(s)=0$ and $X(s)=1$, that is for the absence and presence of GvHD at time s (respectively, dashed and solid lines in Figs. 7 and 8). By looking simultaneously at all the plots in Fig. 8, there is a clear evidence that GvHD affects the cumulative risk of death by increasing it substantially. The fact that the influence of GvHD becomes gradually weaker with later s is a natural consequence of conditioning probabilities on the only individual alive at s. Figure 7 shows instead that the cumulative risk of relapse is not affected by the GvHD, conditionally on

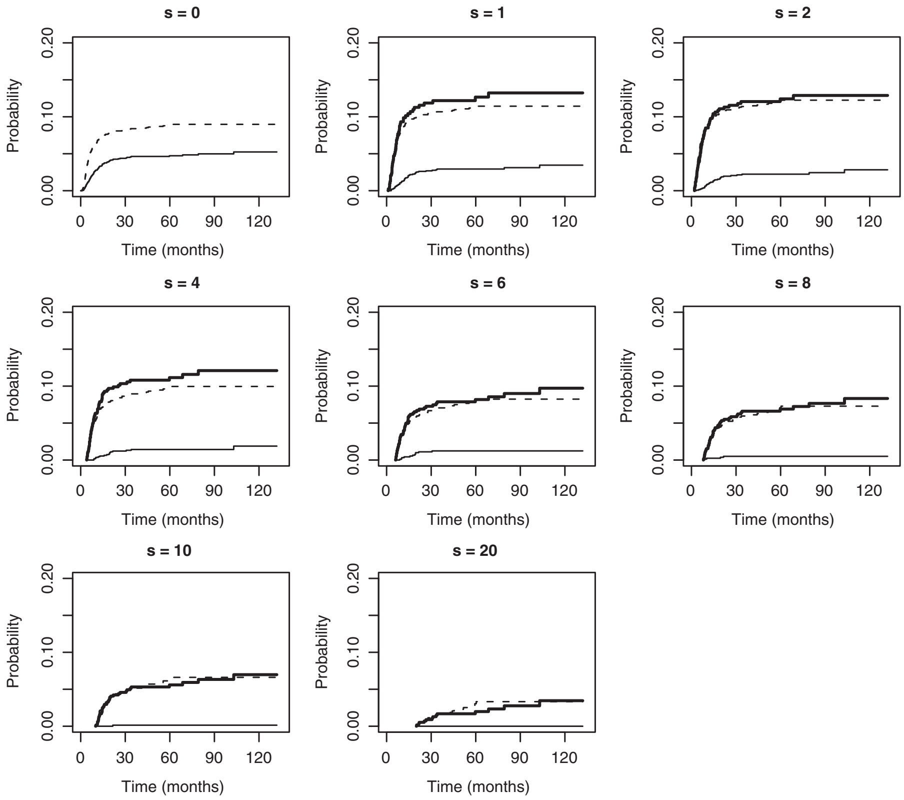

Figure 5 Extended competing risks model (Section 5 and panel (C) of Fig. 1): Estimates of conditional cumulative incidence functions for relapse with or without GvHD. $P_{0,02}(s,t|X(s)=0)$ is estimated by the dashed lines, $P_{0,12}(s,t|X(s)=0)$ by the thin solid lines. $P_{0,12}(s,t|X(s)=1)$ is estimated by the thick solid lines. The probabilities are computed at different landmark times, $s=0,1,2,4,6,8,10,20$. For $s=0$, $P_{0,12}(s,t|X(s)=1)$ is null, since no patients enter the study with GvHD.

being alive at time s for $s>2$, since the curves almost overlap. However, the three plots for $s=0.5,1,2$ might suggest a possible early effect of GvHD on relapse, which vanishes later on after the first 2 months.

The "landmark analysis" provided also a sequence of estimated regression coefficients for GvHD, computed at all times s, which can be useful in suggesting when the GvHD effect on the cause-specific failure rates is stronger over time, and whether it varies substantially. Estimates of those coefficients from our data, summarized in Table 3, indicate that the occurrence of GvHD before s increases the death rate after s more than two fold ($p<0.001$). Since the effect of GvHD on the relapse rate was not significant, we may thus conclude that the presence of GvHD increases (considerably) the future probability of dying after time s. The estimated effects for "age" and "disease" were much the same between times s, and from those previously computed under the other approaches.

Results about the influence of GvHD on the failure risks from the two approaches described in the current subsection, and from the extended competing risks model with an intermediate state, are all in agreement. Of course, a single regression parameter for measuring this influence over time

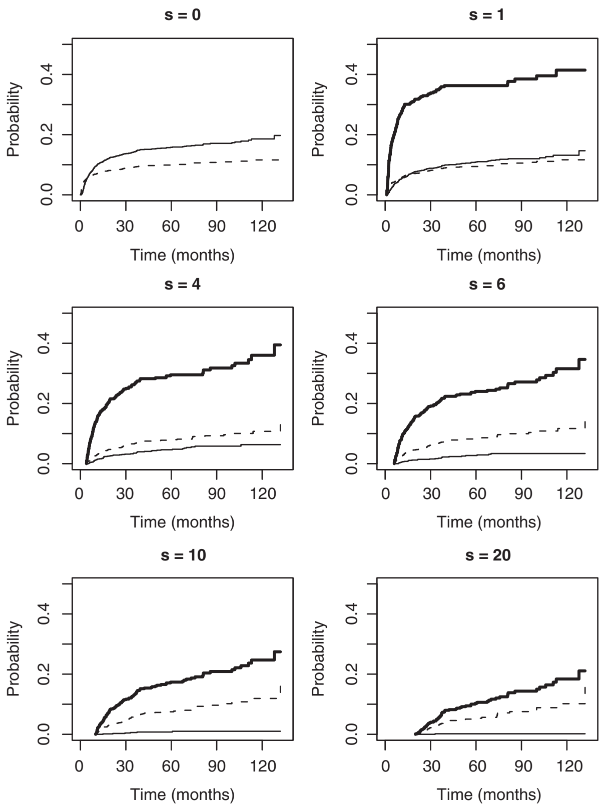

Figure 6 Extended competing risks model (Section 5 and panel (C) of Fig. 1): Estimates of conditional cumulative incidence functions for death with or without GvHD. $P_{0,03}(s,t|X(s)=0)$ are estimated by the dashed lines, $P_{0,13}(s,t|X(s)=0)$ by the thin solid lines. $P_{0,13}(s,t|X(s)=1)$ is estimated by thick solid lines. The probabilities are computed at different landmark times, $s=0,1,2,4,6,8,10,20$. For $s=0$, $P_{0,13}(s,t|X(s)=1)$ is null, since no patients enter the study with GvHD.

would be suitable. On the other hand, the graphical and numerical results we have discussed have provided an exhaustive picture for handling a simple internal covariate, without having to specify a distribution model for this covariate.

7 Discussion

When random (internal) time-dependent covariates are included in the modeling process, it is still possible to estimate cause-specific hazards but prediction of the cumulative incidences and survival probabilities based on these hazards is no longer feasible.

We discussed possible strategies that try to answer this problem within a multi-state model framework. Thus, the cumulative incidences and survival probabilities have been estimated under three different approaches, and they have been compared in order to measure how the internal covariate affects them. However, the two strategies that modify the competing risks model by extending it are only applicable when the internal covariate is categorical while continuous

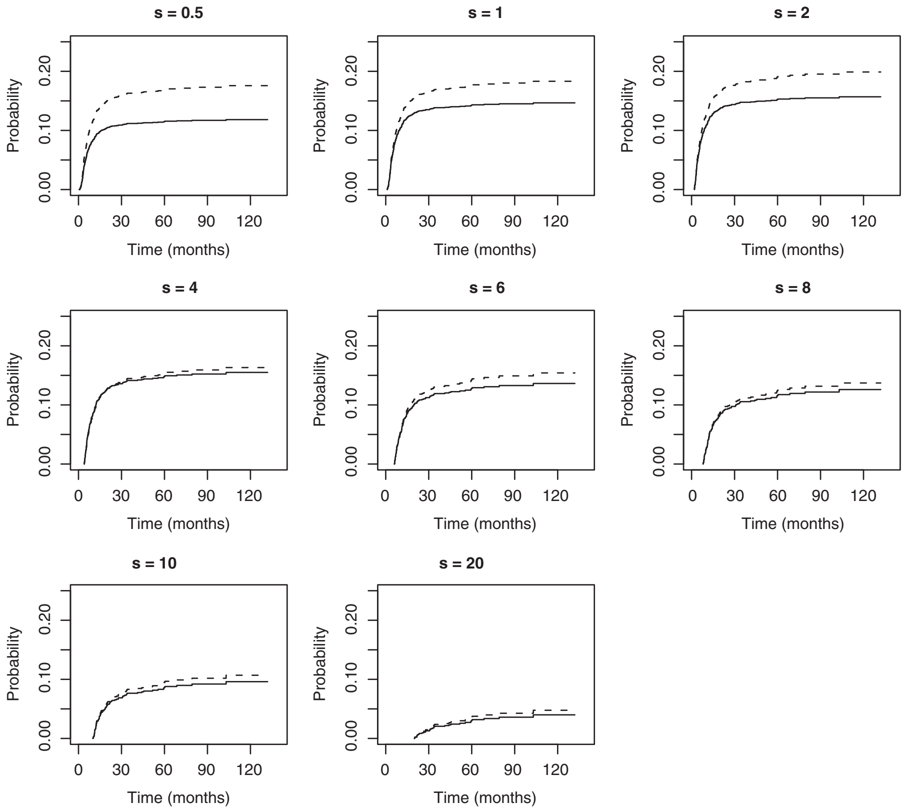

Figure 7 Landmark analysis for the competing risks model: Estimated conditional cumulative incidence functions for relapse. Predictions are for 44-year-old patient with AML disease, both for the case where GvHD is present at time s (solid lines) and for the case of no GvHD at s (dashed lines). Probabilities are computed at the different landmark times $s = 0.5,1,2,4,6,8,10,20$.

covariates would need to be reduced to categorical variables to fit into this framework. In some cases only, this reduction is suitable and will result in a negligible loss of information. Moreover, the extended competing risks model with intermediate states (Section 4), compared with the extended model with enlarged state space (Section 5), requires in addition to specify a model for $X(t)$ and to test for the Markovian assumption.

The third approach presented, which is based on "landmark analysis," is also applicable when internal discrete or continuous covariates are included in the regression models for the cause-specific hazards. Nevertheless, this method accounts only for the information of the internal covariate at the landmark time points.

The application to bone marrow transplant data highlighted the usefulness of the extended competing risks models and the analysis based on landmarks. We estimated the cumulative incidences and survival probabilities easily and studied how the estimated curves changed for different values of the internal covariate (GvHD) over time, and for different time subintervals.

Throughout we have studied regression analysis based on the Cox model. This model allows to simplify the competing risks analysis by imposing the proportionality between the cause-specific hazards. As an example, this assumption can be suitable between α_{1h} and α_{0h}, which then have common regression coefficients for the time-fixed covariates. However, in many situations the

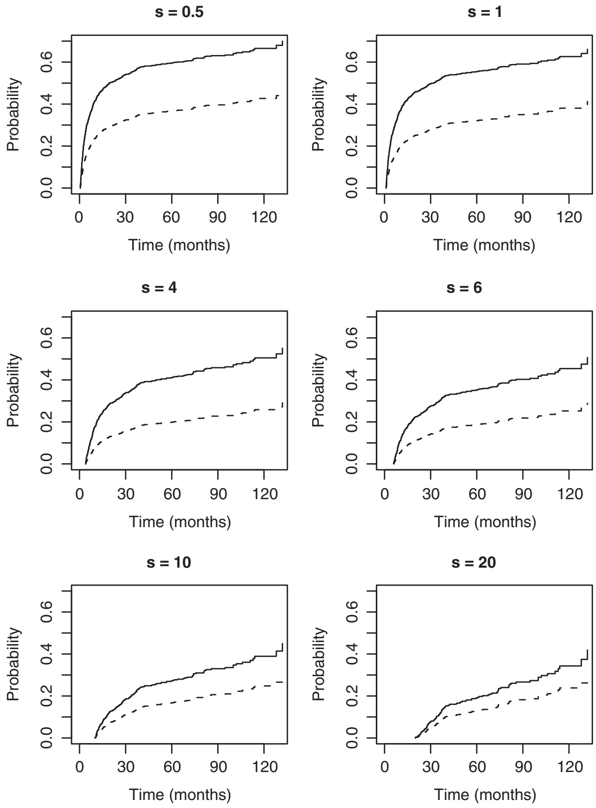

Figure 8 Landmark analysis for the competing risks model: Estimated conditional cumulative incidence functions for death. Predictions are for 44-year-old patient with AML disease, both for the case where GvHD is present at s (solid lines) and for the case of no GvHD at s (dashed lines). Probabilities are computed at the different landmark times s = 0.5,1,2,4,6,8,10,20.

Table 3 Estimates of GvHD effect on the relapse and death rates at different times s, under the Cox model, computed with the landmark analysis.

	Relapse rate			Death rate		
$X(s)$	β	SE(β)	p-Value	β	SE(β)	p-Value
GvHD(1)	−0.007	0.162	0.960	0.759	0.103	<0.001
GvHD(2)	−0.042	0.150	0.780	0.956	0.103	<0.001
GvHD(4)	0.096	0.160	0.550	0.902	0.117	<0.001
GvHD(6)	−0.017	0.181	0.930	0.749	0.129	<0.001
GvHD(10)	−0.044	0.224	0.840	0.551	0.155	<0.001
GvHD(20)	−0.130	0.374	0.730	0.430	0.203	0.034

proportionality assumption might not hold and it is anyway suitable to link the cause-specific hazards to each other. A possible way to obtain this link, relaxing the restrictive proportionality assumption would be to use additive models for the cause-specific hazards. Additive hazards in Markov multi-state regression models were applied to bone marrow transplant data in (Shu and

Klein, 2005). Another possible example is to assume that the cause-specific hazards follow a simple Cox–Aalen model (Scheike and Zhang, 2002), where the baseline hazard depends on a binary time-dependent covariate, as follows

$$\alpha_{12}(t) = [\alpha_{02}^{(0)}(t) + \beta(t)] \exp\{V^T\gamma\}, \qquad \alpha_{02}(t) = \alpha_{02}^{(0)}(t) \exp\{V^T\gamma\}.$$

In this way, the additional effect $\beta(t)$ of the internal covariate, besides the other covariates V, can also vary over time.

A final remark is about our approach concerning the landmark analysis in the competing risks model, explained in Section 5 and applied to the bone marrow transplant data at the end of Section 6. We chose to use a proportional hazards model for each cause-specific hazard when fitting the data and we limited our analysis to separate regression models for the cause-specific hazards (Table 3).

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Conflict of Interests Statement
The authors have declared no conflict of interest.

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